

Demonstration of Resizing and Cropping

```
In [1]: import cv2
import numpy as np
import matplotlib.pyplot as plt

from resizing_64 import resizing_to_64
from cropp_face import cropping_face
from load_datasets import CustomDatasetLoaderwithoutResizing
from skip_image import SkipImage
```

```
In [2]: label_map = {
    1: "surprise",
    2: "fear",
    3: "disgust",
    4: "happy",
    5: "sad",
    6: "angry"
}

loader = CustomDatasetLoaderwithoutResizing(csv_path="datasets.csv")
images, labels = loader.load_dataset() #I have not used the max argument due to the small size of the test
```

Resizing the images

```
In [3]: for i in range(5):
    image = images[i]
    label = labels[i]

    w, h, c = image.shape

    plt.tight_layout()
    ax = plt.subplot(1, 5, i + 1)
    ax.imshow(image)
    ax.set_title(f"size: {w}x{h} \n label: {label_map[label]}")
    ax.axis("on")
```

```
plt.show()
```



```
In [4]: resizer = resizing_to_64()
```

```
for i in range(5):

    image = images[i]
    label = labels[i]

    resized_image = resizer.process(image)
    w, h, c = resized_image.shape

    plt.tight_layout()
    ax = plt.subplot(1, 5, i + 1)
    ax.imshow(resized_image)
    ax.set_title(f"size: {w}x{h} \n label: {label_map[label]}")
    ax.axis("on")

plt.show()
```



Crop

```
In [8]: for i in range(20):
        image = images[i]
        label = labels[i]

        plt.tight_layout()
        ax = plt.subplot(5, 4, i + 1)
        ax.imshow(image)
        ax.set_title("Not cropped")
        ax.axis("off")

plt.show()
```

Not cropped



Not cropped



Not cropped



Not cropped



Not cropped



Not cropped



Not cropped



Not cropped



Not cropped



Not cropped



Not cropped



Not cropped



Not cropped



Not cropped



Not cropped



Not cropped



Not cropped



Not cropped



Not cropped



Not cropped



```
In [12]: #Cropped
face_cropper = cropping_face()
skipped = 0

for i in range(20):
    image = images[i]
    label = labels[i]

    w, h, c = image.shape

    try:

        transformed_images = face_cropper.process(image)
```

```
plt.tight_layout()
ax = plt.subplot(5, 4, i + 1)
ax.imshow(transformed_images)

ax.axis("off")
except SkipImage:
    skipped += 1
    continue

plt.show()
print("Skipped images:", skipped)
```



Skipped images: 4